



Question 1

A 25 kg box is dragged on a rough horizontal x-y plane from $\vec{r}_1 = (1.3 \vec{i} + 1.4 \vec{j})\text{m}$ to $\vec{r}_2 = (8.3 \vec{i} + 2.9 \vec{j})\text{m}$ by a force $\vec{F} = (4 \vec{i} - 6 \vec{j})$. How much work is done by the force?

- A. 19.0 J
- B. 37.0 J
- C. 15.8 J
- D. -75.6 J

Question 2

Two resistors with resistance of 3Ω and 6Ω are connected in parallel, and their combination is connected in series to a 4Ω resistor. If the overall combination of the resistors is connected to a 12 V source, how much is the voltage drop across the 6Ω resistor?

- A. 8 V
- B. 4 V
- C. 2 V
- D. 12 V

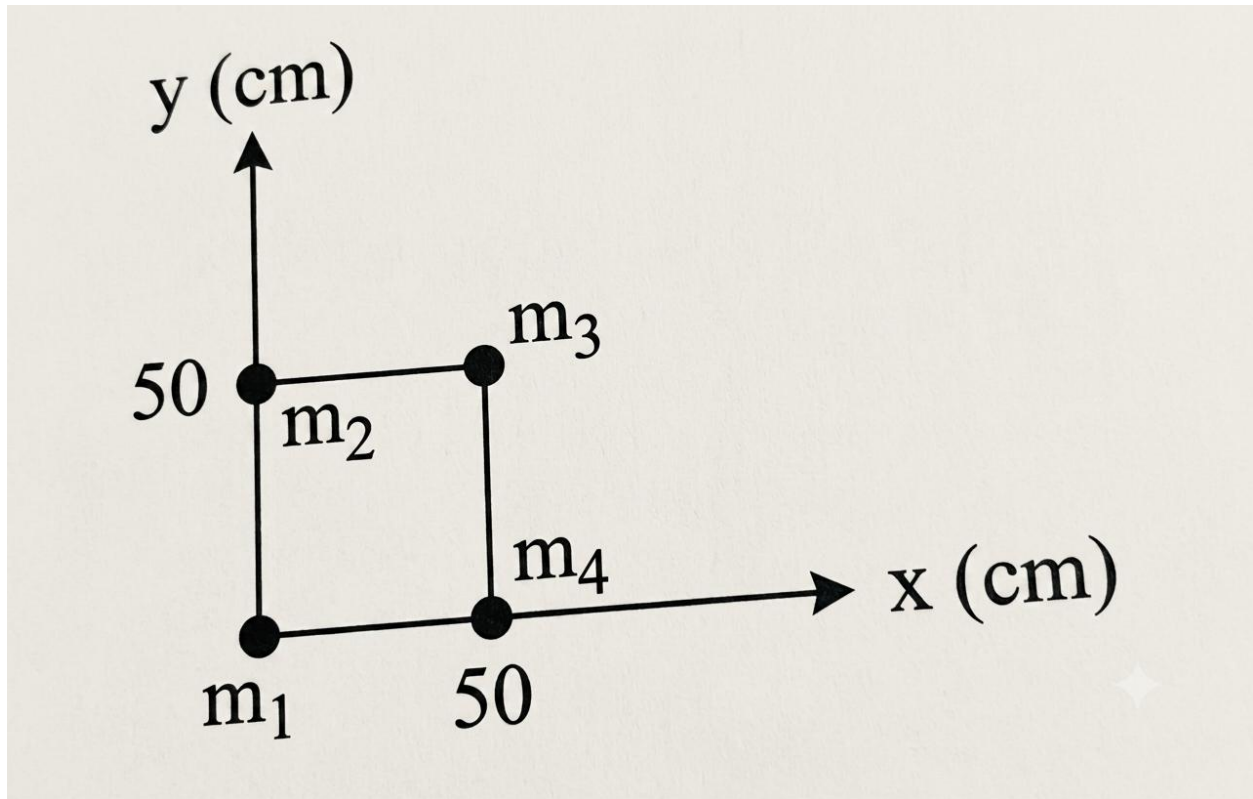
Question 3

A constant force of 250.0 N is applied horizontally on a box that lays on a horizontal frictionless surface. If the box accelerates with 4.0 m/s^2 , what is its mass?

- A. 125.0 kg
- B. 41.7 kg
- C. 10.4 kg
- D. 62.5 kg

Question 4

Four masses are placed on the corners of a square on the x-y plane as shown in the figure below, where $m_1 = m_2 = 200 \text{ g}$ and $m_3 = m_4 = 500 \text{ g}$. Where is the center of mass (x_{cm}, y_{cm}) of the system?



- A. (25,37.14) cm
- B. (25,25) cm
- C. (37.14,50) cm
- D. (35.71,25) cm

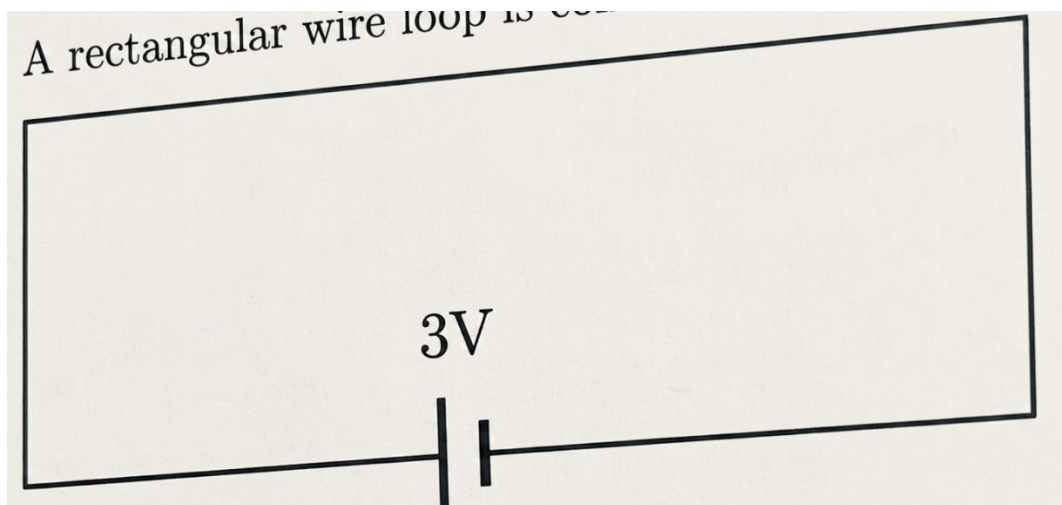
Question 5

Ethiopian Physical Society is established by interested Ethiopian physics professionals to:

- A. Prepare physics curriculum under the ministry of education directives.
- B. Contribute its own share to promote physics education and research.
- C. Take part in election and participate in government formation.
- D. Monitor and supervise the curriculum implementation in Ethiopian high schools.

Question 6

A rectangular wire loop is connected to a 3 V battery as shown below. What will happen to the direction of the magnetic field inside the loop if the terminals of the battery are reversed?



- A. It will change from into the plane to out of the plane.
- B. It will change from out of the plane to into the plane.
- C. It will change from counterclockwise to clockwise.
- D. It will change from clockwise to counterclockwise.

Question 7

The magnetic field created by a current-carrying wire at a point from the wire will:

- A. Increase exponentially with the increase in the magnitude of the current.
- B. Decrease linearly with the increase in the magnitude of the current.
- C. Decrease exponentially with the increase in the magnitude of the current.
- D. Increase linearly with the increase in the magnitude of the current.

Question 8

The iodine isotope ^{131}I , which has an eight-day half-life, is used in nuclear medicine. What is the decay constant of the isotope? ($\ln 2 = 0.693$)

- A. 5.64 day^{-1}
- B. $8.66 \times 10^{-2} \text{ day}^{-1}$
- C. $1.17 \times 10^{-2} \text{ day}^{-1}$
- D. 12.5 day^{-1}

Question 9

The velocity of car B with respect to a passenger in car A is 80 km/hr towards West. If the velocity of car A with respect to the ground is 30 km/hr towards East, what would be the velocity of car B with respect to the ground?

- A. 50 km/hr towards West
- B. 50 km/hr towards East
- C. 110 km/hr towards East
- D. 110 km/hr towards West

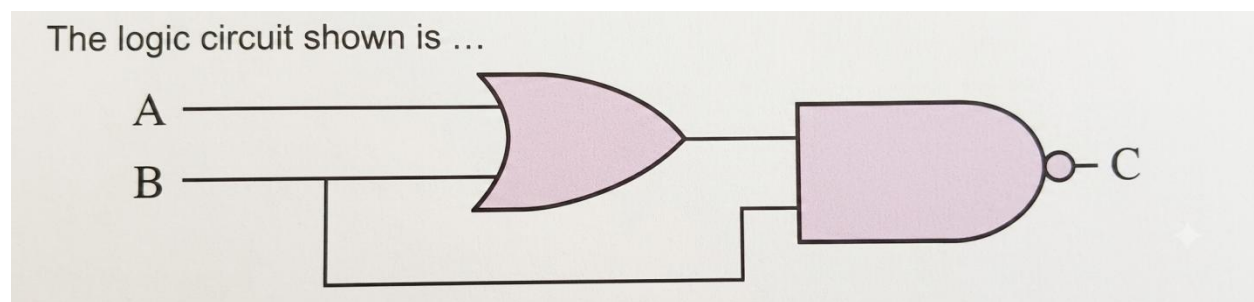
Question 10

Which one of the following statements correctly describes the rotation of an object around a fixed axis?

- A. Angular acceleration is perpendicular to the plane of rotation and in the same direction as angular velocity if the rotation rate is increasing.
- B. Particles in the object that are closer to the axis will have different angular displacement from particles that are far from it.
- C. Particles in the object closer to the axis will have larger angular velocity than particles that are farther from the axis.
- D. Particles that are closer to the axis will have similar translational displacement from particles that are far from it.

Question 11

The logic circuit shown below consists of OR and NAND gates. Which of the following tables represents the correct truth table for the circuit?



Select the correct truth table for the circuit:

A.

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

B.

A	B	C
0	0	0
0	1	1
1	0	0
1	1	1

C.

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

D.

A	B	C
0	0	1
0	1	0
1	0	1
1	1	0

Question 12

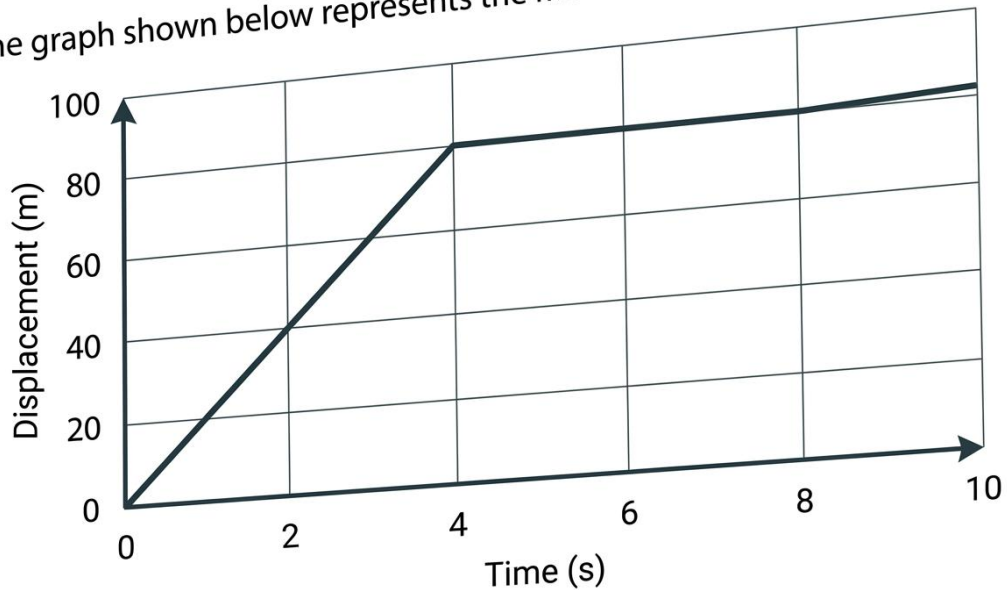
Which one of the following sets contains only primary colors?

- A. {red, blue, orange}
- B. {blue, green, red}
- C. {green, blue, black}
- D. {red, green, yellow}

Question 13

The graph shown below represents the motion of a car along a straight line. What are the magnitudes of the velocities of the car for the first 4 seconds and the last 4 seconds, respectively?

The graph shown below represents the motion of a car along a straight line.

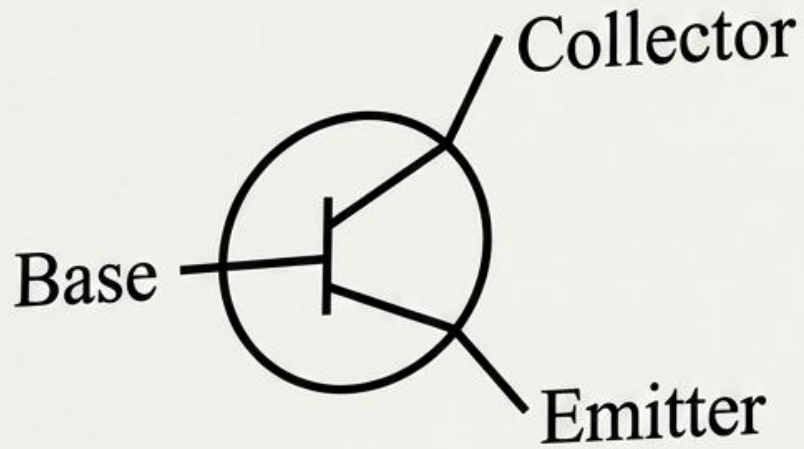


- A. 0 m/s and 20 m/s
- B. 20 m/s and 0 m/s
- C. 20 m/s and 20 m/s
- D. 10 m/s and 0 m/s

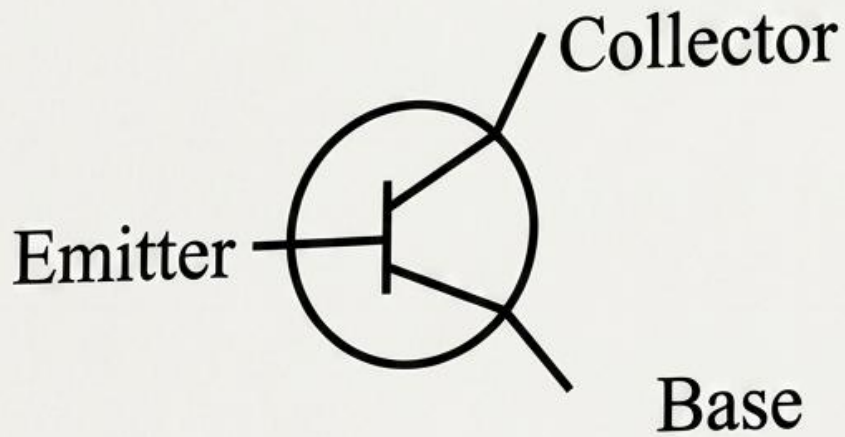
Question 14

Which of the following is a correct schematic symbol for an NPN transistor with its labelled terminals?

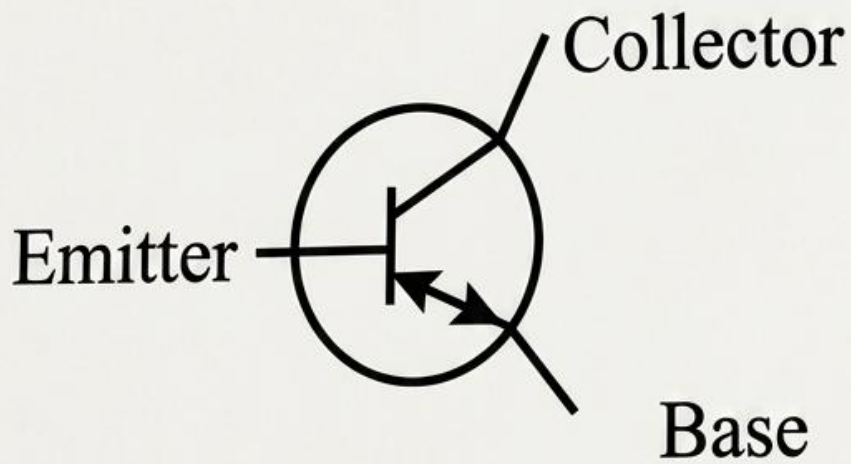
☐ A.



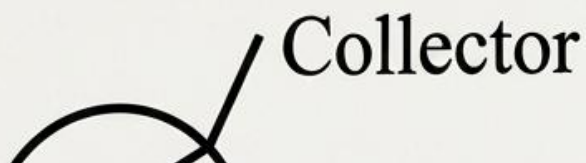
☐ B.



☐ C.



☐ D.



Question 15

A water hose of diameter 4 cm is connected to a tap of diameter 6 cm. The water flow rate from the tap is $1.57 \times 10^{-4} \text{ m}^3/\text{s}$. What will be the speed of the water in the hose?

- A. 0.125 m/s
- B. 0.393 m/s
- C. 0.174 m/s
- D. 0.560 m/s

Question 16

Suppose you want to check, without any touch, if there is a current flowing through a wire or not. Which one of the following devices is used?

- A. Test light
- B. Voltmeter
- C. Compass
- D. Ohmmeter

Question 17

Which one of the following statements describes the characteristics of a third-order lever?

- A. The load is between the effort and fulcrum.
- B. The fulcrum is between the load and the effort.
- C. The effort distance is greater than load distance.
- D. The effort is between the load and the fulcrum.

Question 18

Which one of the following statements correctly describes the characteristics of equipotential lines?

- A. For a point charge, the equipotential lines are straight lines and parallel to each other.
- B. For a uniform electric field, the equipotential lines are parallel to that of the field lines.
- C. Electric potential difference is non-zero between two points on different equipotential lines.
- D. Electric potential difference is non-zero between two points on the same equipotential line.

Question 19

A strong nuclear force is:

- A. An attractive force between the nucleus and the electrons around the nucleus.
- B. A short-range attractive force that acts between nucleons in the nucleus.
- C. A short-range repulsive force that acts between protons in the nucleus.
- D. A short-range force which acts inside the individual nucleons, and can split a neutron into a proton and electron.

Question 20

Consider two objects A and B, where A has twice the mass of B. If object B is raised to a height h above the ground and object A is raised to a height $h/2$, which of the following is correct about their gravitational potential energies?

- A. The gravitational potential energy of A is equal to that of B.
- B. The gravitational potential energy of A is twice that of B.
- C. The gravitational potential energy of B is twice that of A.
- D. The gravitational potential energy of A is 4 times that of B.

Question 21

A quantity that describes deformation of materials that is given by fractional change in length or volume is:

- A. Stress
- B. Strain
- C. Elasticity
- D. Density

Question 22

Electric field at a point is defined as the electric:

- A. Force per unit mass that a test charge experiences when placed at that point.
- B. Force that a positive test charge exerts on another point charge placed at that point.
- C. Potential energy per unit charge of a test charge placed at that point.
- D. Force per unit charge that a positive test charge experiences when placed at that point.

Question 23

A current of 6 A flows in a hot plate when the potential difference between the terminals is 102 V. How much current will flow if the potential difference is reduced to 68 V for a plate that obeys Ohm's law?

- A. 2 A
- B. 4 A
- C. 6 A
- D. 3.4 A

Question 24

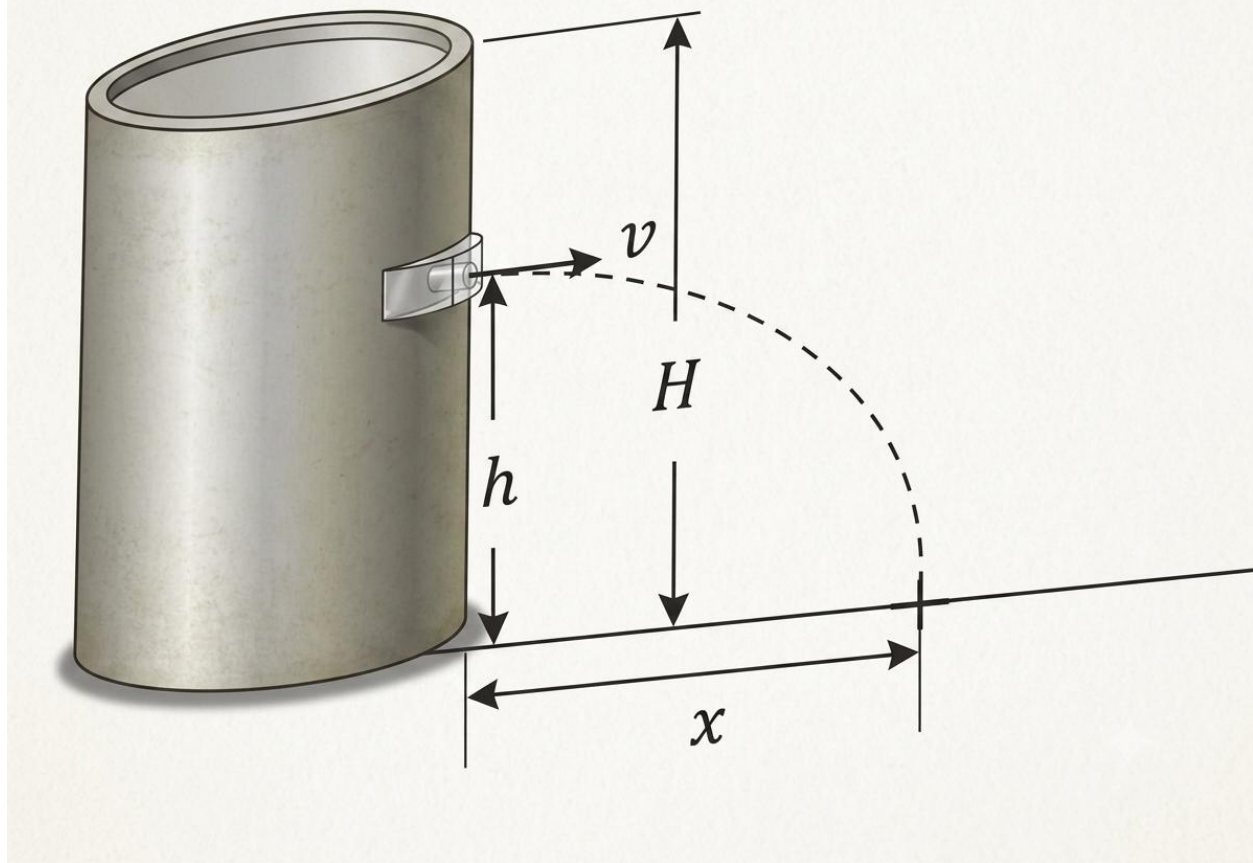
A car is moving with a constant acceleration. This means that:

- A. Its velocity changes equally with equal intervals of time.
- B. It moves with constant speed.
- C. It moves with constant velocity.
- D. Its position changes equally with equal intervals of time.

Question 25

A barrel of height H is fully filled with water. A bullet shoots the barrel at height h above the ground and the water is projected at a horizontal speed of v as shown in the figure below. At what horizontal distance (x) does the spilled water strike the ground?

on projectile



- A. $x=2(H-h)/g$
- B. $x=v\sqrt{g}$
- C. $x=g/H$
- D. $x=v\sqrt{2h/g}$

Question 26

A collision between two particles is known as:

- A. Elastic if the total kinetic energy is not conserved but the total momentum is conserved.
- B. Elastic if both the total kinetic energy and momentum are conserved.
- C. Inelastic if the total kinetic energy is conserved but the total momentum is not conserved.
- D. Inelastic if both the total kinetic energy and momentum are conserved.

Question 27

A parallel-plate capacitor has square metallic plates each of area 100 cm^2 separated by 1 mm . The space between the plates is filled by a dielectric of dielectric constant $k=4$. If the plates are connected to the terminals of a 12 V battery, how much charge is stored on the capacitor?

- A. 360 nC
- B. 425 nC
- C. 432 nC

D. 1.08 nC

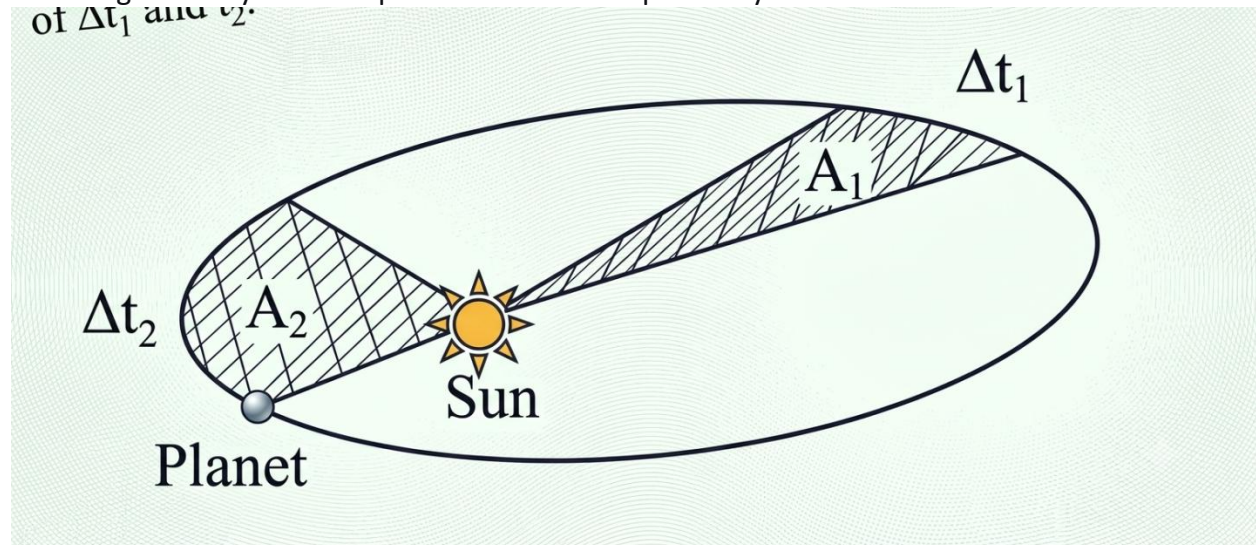
Question 28

When an n-type and p-type semiconductors form a junction, it will form a diode:

- A. That conducts significant current in both forward and reverse biasing.
- B. That will have an ohmic type current-voltage characteristics.
- C. Where a p-side is anode and the n-side is cathode at the junction.
- D. That conducts current in the reverse biasing but not in the forward biasing.

Question 29

A certain planet revolving around the sun sweeps areas A_1 and A_2 at different positions during the time intervals of Δt_1 and Δt_2 respectively as shown in the figure below. Which one of the following correctly states Kepler's second law of planetary motion?
of Δt_1 and Δt_2 .



- A. If $\Delta t_1 > \Delta t_2$, then $A_1 = A_2$
- B. If $\Delta t_1 = \Delta t_2$, then $A_1 = A_2$
- C. If $\Delta t_1 > \Delta t_2$, then $A_1 < A_2$
- D. If $\Delta t_1 = 2\Delta t_2$, then $A_1 = (A_2)^2$

Question 30

Co-planar vectors are vectors which are:

- A. Lying in the same plane.
- B. Perpendicular and lying in two parallel planes.
- C. Parallel and lying in two parallel planes.
- D. Lying in different planes.

Question 31

A packed books of mass 3 kg is lying on the surface of a 0.90 m high table. If you pick it up and place it on a bookshelf 2.0 m above the floor, the work done by you and by gravity, respectively, are:

- A. 60 J and 60 J
- B. 33 J and 33 J
- C. 33 J and 33 J

D. 60 J and -60 J

Question 32

Which one of the following statements does NOT correctly explain the contribution of technology to the advancement of physics?

- A. The invention of space shuttles enabled experiments in space.
- B. The development of light detectors enabled spectral analysis.
- C. The invention of x-rays enabled proper medical imaging.
- D. The development of x-rays enabled study of atomic structure.

Question 33

For an athlete who runs on a track for a specified time interval, her distance and displacement can be different in that distance:

- A. Can be negative whereas displacement cannot be.
- B. Is less than the magnitude of displacement.
- C. Can be zero whereas the displacement cannot be.
- D. Is greater than the magnitude of displacement.

Question 34

For a person with the vision defect known as hypermetropia:

- A. Close objects cannot be clearly seen, so the defect can be corrected by concave lenses.
- B. Far objects cannot be clearly seen, so the defect can be corrected by convex lenses.
- C. Far objects cannot be clearly seen, so the defect can be corrected by concave lenses.
- D. Close objects cannot be clearly seen, so the defect can be corrected by convex lenses.

Question 35

Given two parallel conductors C_1 and C_2 carrying currents in the same direction as shown in the figure below. The force between the two conductors is attractive because the:



- A. Magnitude of the magnetic field produced by I_1 at C_2 is the same as that produced by I_2 at C_1 .
- B. Direction of the magnetic field produced by I_1 at C_2 is opposite to that produced by I_2 at C_1 .

- C. Magnitude of the magnetic field produced by I_1 at C_2 is different from that produced by I_2 at C_1 .
- D. Direction of the magnetic field produced by I_1 at C_2 is the same as that produced by I_2 at C_1 .

Question 36

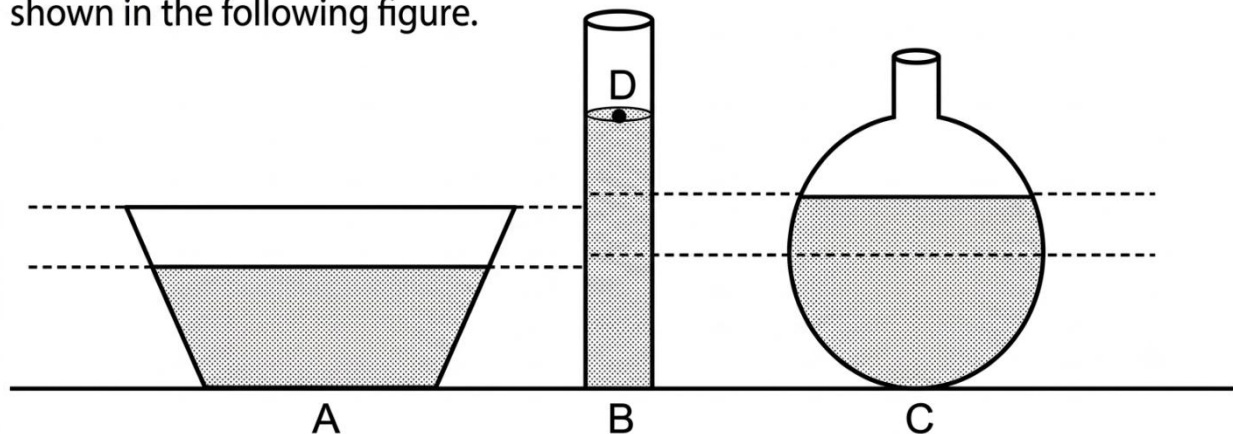
A square wire loop with a side length of 12 cm is placed in such a way that its area vector makes an angle of 60° with a magnetic field of 10 T. If the loop is slowly pulled out of the magnetic field in 6 s, how much electromotive force is induced in the loop during this time?

- A. 24 mV
- B. 17.2 mV
- C. 12 mV
- D. 20.9 mV

Question 37

Three different shapes of containers are filled with water at different levels and opened to the atmosphere as shown in the following figure. At which point does a student measure the highest pressure?

shown in the following figure.



- A. At point A
- B. At point B
- C. At point C
- D. At point D

Question 38

Lenses form images by refraction of light rays from the object such that the position, size, and nature of the image is determined from a point where the refracted rays or their extensions meet. After refraction from a lens, a light ray:

- A. Through the optical center will emerge parallel to the principal axis.
- B. Through the principal focus will emerge parallel to the principal axis.
- C. Parallel to the principal axis will emerge without any deviation.
- D. Parallel to the principal axis will emerge through the optical center.

Question 39

The top end of a 6 m long and $3 \times 10^{-6} \text{ m}^2$ cross-sectional area wire is attached to a ceiling, and the wire is suspended vertically downward. If a 120 kg mass is attached to the bottom end of the wire, and the wire's Young's modulus is $6 \times 10^{10} \text{ N/m}^2$, what will be the change in length?

- A. 0.05 m
- B. 0.04 m
- C. 0.06 m
- D. 0.02 m

Question 40

The speed of sound in air at 0°C is 331.0 m/s. What is the approximate speed of sound in air at 27°C ?

- A. 346.4 m/s
- B. 358.0 m/s
- C. 434.9 m/s
- D. 331.0 m/s

Question 41

If the coefficients of thermal expansion of a material are designated by α for linear expansion, β for surface (area) expansion, and γ for volume expansion, these parameters are related as:

- A. $\gamma = 2\alpha$
- B. $2\beta = 3\gamma$
- C. $3\beta = 2\gamma$
- D. $\beta = 3\alpha$

Question 42

The amount of electric charge on a particle or object is an integral multiple of the charge of an electron. This property indicates:

- A. Charge creation.
- B. Charge induction.
- C. Charge quantization.
- D. Charge conservation.

Question 43

Which of the following is true about anechoic images of tissues in ultrasound imaging? They appear on the screen as:

- A. Black because the tissues do not reflect any sound wave.
- B. Light gray because the tissues do not reflect any sound wave.
- C. Black because the tissues reflect intense sound waves.
- D. Light gray because the tissues reflect intense sound waves.

Question 44

A stream of water from a wide river enters a narrower and shallow gate of a lake. Which one of the following statements is true?

- A. The speed of the water decreases as it enters the gate of the lake.
- B. The speed of the water increases as it enters the gate of the lake.

- C. The density of the water entering the lake is higher than the water leaving the river.
- D. The volume of the water entering the lake is higher than the water leaving the river.

Question 45

A light ray passes from a layer of benzene of refractive index $(n=1.50)$ to water of refractive index $(n=1.33)$. If the angle of incidence is 37° , what will be the angle of refraction?

- A. $\sin^{-1}(0.90)$
- B. $\sin^{-1}(0.80)$
- C. $\sin^{-1}(0.53)$
- D. $\sin^{-1}(0.68)$

Question 46

Consider a car performing a motion with constant acceleration (a) starting from position x_0 at time $t=0$ with initial velocity v_0 . If v and x stand for velocity and position at time t , which one of the following equations correctly governs the motion of the car?

- A. $x = -x_0 + \frac{2(v+v_0)}{t}$
- B. $x = x_0 + \left(\frac{v+v_0}{2}\right)t$
- C. $x = x_0 + \frac{v^2+v_0^2}{2a}$
- D. $x = x_0 + \left(\frac{v-v_0}{2}\right)t$

Question 47

If an arrow is drawn to a scale to describe a given vector, then its:

- A. Length corresponds to the vector magnitude.
- B. Tail stands for the terminal point of the vector.
- C. Length corresponds to the vector direction.
- D. Head represents the initial point of the vector.

Question 48

From the given lists of machines, which one contains only simple machines?

- A. Gas-cylinder, Wedge, Screw, Lever
- B. Compound pulley, Wheel and axle, Seat belt, Inclined plane
- C. Washing machine, Car engine, Wedge, Lever
- D. Inclined plane, Screw, Wheel and axle, Wedge

Question 49

A traveling wave has a speed v , wavelength λ , period T , and frequency f . Which equation correctly states the wave equation?

- A. $v=\lambda T$
- B. $f=1/(\lambda v)$
- C. $v=\lambda f$

D. $T=1/(\lambda f)$

Question 50

A 200 g piece of ice at 0°C is placed in a certain amount of water at 20°C . This system is in a container of negligible heat capacity and is insulated from its surroundings. How much should be the minimum mass of water to melt all the ice at 0°C to 0°C temperature? (The latent heat of fusion of ice is $L_f=3.33\times 10^5 \text{ J/kg}$)

- A. 666 g
- B. 200 g
- C. 400 g
- D. 793 g

Question 51

Conservative force differs from the nonconservative one in that, the work done by:

- A. Conservative force along a closed path is non-zero unlike that by non-conservative force.
- B. Conservative force is path independent unlike that by non-conservative one.
- C. Non-conservative force is path independent unlike that by the conservative one.
- D. Non-conservative force along a closed path is zero unlike that by non-conservative force.

Question 52

Starting from rest, a car accelerates uniformly for 400 m, then continues for 600 m with constant velocity. It then decelerates uniformly with 1 m/s^2 until it stops after traveling for 200 m. What is the average velocity for the total trip?

- A. 20.0 m/s
- B. 6.5 m/s
- C. 13.3 m/s
- D. 10.0 m/s

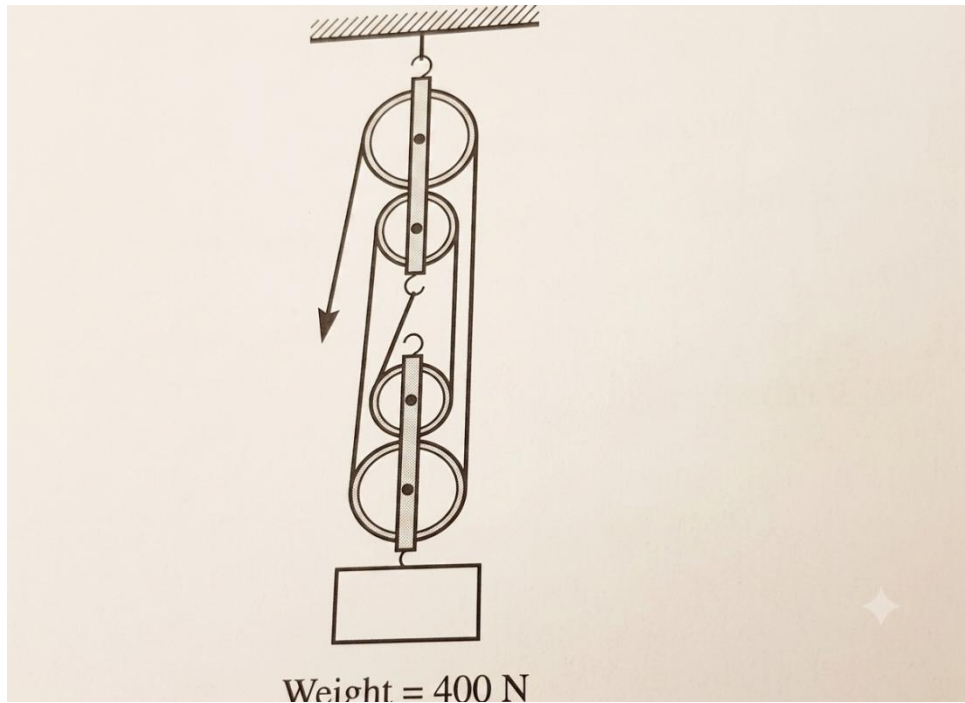
Question 53

If a student walks 100 m due 37° North of East and then 120 m towards South, what is the displacement moved by the student?

- A. 220 m 53° North of East
- B. 100 m 53° North of East
- C. 220 m 37° South of East
- D. 100 m 37° South of East

Question 54

A compound pulley system consisting of two movable and two fixed pulleys is used to lift a 400 N object as shown in the figure below. If the frictional force between the pulleys and the rope is negligible, what is the mechanical advantage of the pulley system?



- A. 4
- B. 2
- C. 5
- D. 1

Question 55

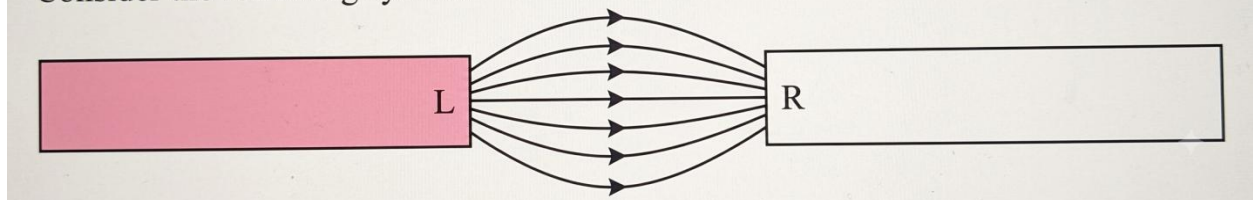
The upward force that an object immersed in a fluid experiences is known as:

- A. Weight of the object.
- B. Turbulent force.
- C. Buoyant force.
- D. Force of the fluid.

Question 56

Consider the magnetic field lines between two bar magnets are partially drawn as shown in the figure below. The two poles of the magnets indicated by R and L respectively are:

Consider the following system:



- A. North and South poles.
- B. North and North poles.
- C. South and North poles.
- D. South and South poles.

Question 57

Which one of the following statements describes heat?

- A. It is the form of energy that transfers when two objects of different temperatures are brought in thermal contact.
- B. It is the form of energy which is stored in an object and increases when the object's temperature increases.
- C. It is a property of a physical system which is related to the average kinetic energy of the particles of the system.
- D. It is a form of energy that transfers between two systems by doing work in the process to achieve mechanical equilibrium.

Question 58

To determine the specific heat capacity of a solid, we can use a method called:

- A. Mixture method with heat energy supplied to the solid for a known time by an electrical heater of known power.
- B. Electrical method with the heat energy supplied for a known time by an electrical heater of known power.
- C. Electrical method with heat energy supplied to the solid by mixing hot water and solid in a calorimeter.
- D. Mixture method with the heat energy supplied to the solid for unknown time by an electrical heater of known power.

Question 59

Which one of the following statements correctly describes a uniform circular motion of a particle in a horizontal plane? Its:

- A. Centripetal acceleration is a non-zero constant.
- B. Tangential velocity is a non-zero constant.
- C. Angular velocity is a non-zero constant.
- D. Tangential acceleration is a non-zero constant.

Question 60

Lenses form images by refraction of light rays from the object such that the position, size, and nature of the image is determined from a point where the refracted rays or their extensions meet. After refraction from a lens, a light ray:

- A. Through the optical center will emerge parallel to the principal axis.
- B. Through the principal focus will emerge parallel to the principal axis.
- C. Parallel to the principal axis will emerge without any deviation.
- D. Parallel to the principal axis will emerge through the optical center.

ANSWER KEY FOR 60 PHYSICS QUESTIONS

Question	Answer
1	A

Question	Answer
2	B
3	D
4	D
5	B
6	A
7	D
8	B
9	A
10	A
11	D
12	B
13	B
14	A
15	A
16	C
17	D
18	C
19	B

Question	Answer
20	A
21	B
22	D
23	B
24	A
25	D
26	B
27	B
28	C
29	B
30	A
31	B
32	C
33	D
34	D
35	B
36	C
37	B

Question	Answer
38	B
39	B
40	A
41	C
42	C
43	A
44	B
45	D
46	B
47	A
48	D
49	C
50	D
51	B
52	C
53	D
54	A
55	C

Question	Answer
56	A
57	A
58	B
59	C
60	B



THE END
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